

On the Maximum Switching Number to Obtain Non-crossing Convex Cycles*

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Abstract

Let S be a set of $n \geq 3$ points arranged in convex position in the plane. A convex cycle is a Hamiltonian cycle that connects all points of S with straight line segments. The switching number of a convex cycle C is the minimum number of steps required to transform C into a non-crossing convex cycle using the operation that deletes two crossing edges, adds two other edges and preserves hamiltonicity. Let $\delta(n)$ be the maximum switching number amongst all possible convex cycles on S . Recently, Oda and Watanabe in [Y. Oda, M. Watanabe, *Lecture Notes in Computer Science*, Vol. 4535, pp. 155-165] showed that $n-2 \leq \delta(n) \leq 2n-7$ for $n \geq 7$ and conjectured that the exact value of $\delta(n)$ is $n-2$ when $n \geq 7$. In this paper, we give an affirmative answer for this conjecture.

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1. Introduction

Let S be a set of $n \geq 3$ points in convex position in the plane and suppose that all points of S are labeled from 1 to n in clockwise direction. A convex cycle is a Hamiltonian cycle (i.e., a spanning cycle of S) whose edges connect all points of S with straight line segments. We denote a convex cycle passing through points $p_1, p_2, \dots, p_n$ by $(p_1, p_2, \dots, p_n)$ and regard the cycles $(p_1, p_2, \dots, p_n)$, $(p_2, \dots, p_n, p_1)$ and $(p_n, p_{n-1}, \dots, p_1)$ to be equivalent. Two edges e and e' are said to *cross* each other if they have an interior point in common. In such a case, $\{e, e'\}$ is called a pair of *crossing edges*. For a given convex cycle $C = (p_1, p_2, \dots, p_n)$, a *flip* is a replacement of a pair of crossing edges $e = (p_{i-1}, p_i)$ and $e' = (p_j, p_{j+1})$ in C by the edges $f = (p_{i-1}, p_j)$ and $f' = (p_i, p_{j+1})$ (provided that f and f' were not present in C) while preserves hamiltonicity. We write $C \xrightarrow{e, e'} C'$ to denote such an operation. Accordingly, the flip can also be considered to reverse a subpath from p_i to p_j on C . For example, Fig. 1 shows a convex cycle $(1, 3, 5, 4, 7, 2, 8, 6)$ that applies the flip on the pair of crossing edges

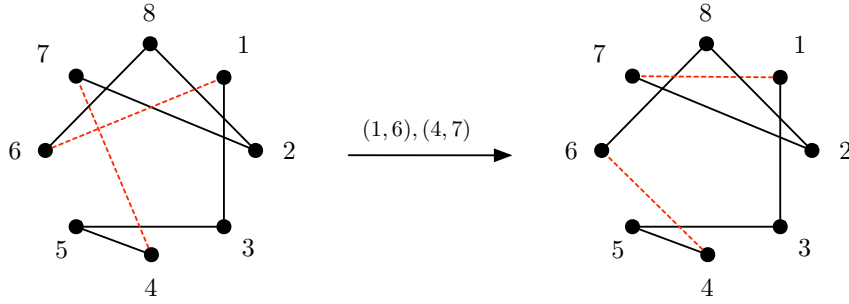


Figure 1: A flip applies to a convex cycle.

$(1, 6)$ and $(4, 7)$.

Let $\mathcal{C}_n$ be the set of convex cycles on S of order n and denote the unique non-crossing convex cycle as $I_n = (1, 2, 3, \dots, n)$. For each convex cycle $C \in \mathcal{C}_n$, the *switching number* of C , denoted by $\delta(C)$, is the minimum number of flips required to transform C into I_n . Oda and Watanabe [12] recently considered the following problem: Find the value of $\delta(n)$ that is defined by

$$\delta(n) = \max_{C \in \mathcal{C}_n} \delta(C).$$

Then, the bounds of $\delta(n)$ in that paper were given as follows:

Theorem 1 (see [12].) *For any integer $n \geq 4$, $\delta(n) \leq 2n - 7$.*

Theorem 2 (see [12].) *For any integer $n \geq 3$,*

$$\delta(n) \geq \begin{cases} n - 3 & \text{if } n \in \{3, 4, 6\}, \\ n - 2 & \text{if } n = 5 \text{ or } n \geq 7. \end{cases}$$

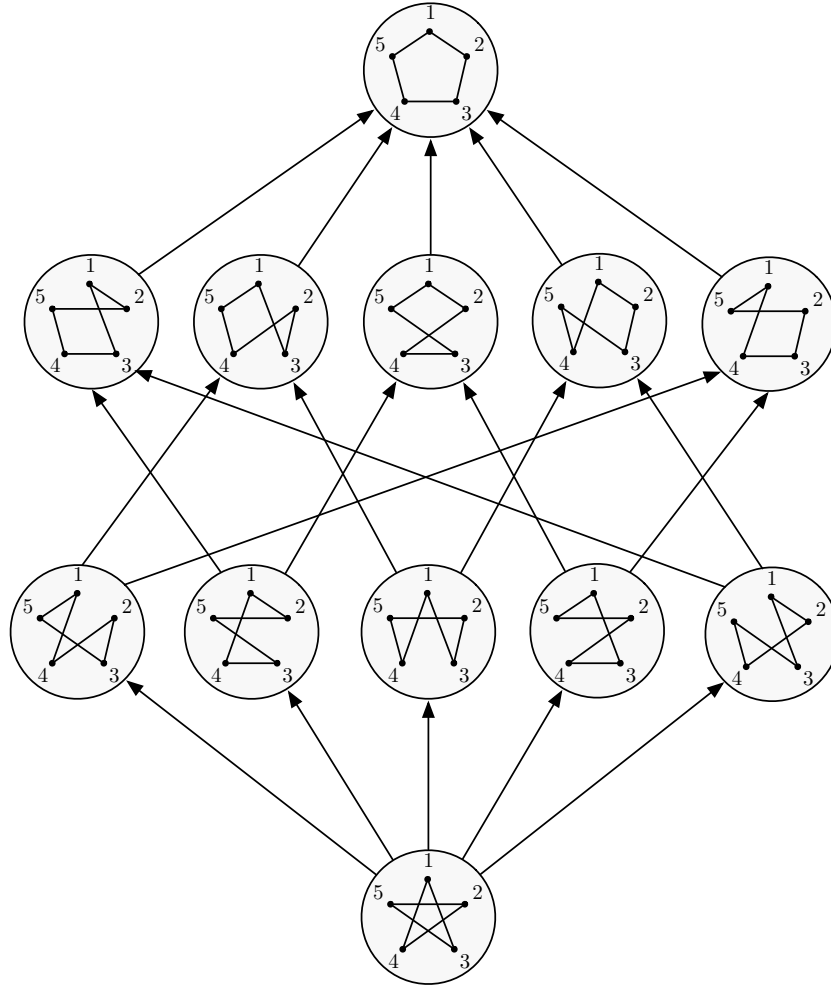
Note that Oda and Watanabe [12] also pointed out that the lower bounds are tight for $3 \leq n \leq 6$. Moreover, by providing some computational results of $\delta(n)$ for integer n from 7 to 10, they conjectured that $\delta(n) = n - 2$ for $n \geq 7$. The aim of this paper is to show that the conjecture is in the affirmative.

A relevant research in this area has been considered by Hagita et al. [10] to deal with

the computation of the diameter of a so-called transition graph whose vertices correspond to those convex cycles, and two vertices are adjacent if their corresponding convex cycles can be transformed each other by exchanging two edges and without concerning about the deleted edges to be crossing or non-crossing. This inspired us with a similar definition. Formally, the *transition graph* $\mathbb{G}_n$ is a digraph consisting of all convex cycles of $\mathcal{C}_n$ as its vertices and an arc goes from a convex cycle C to another convex cycle C' if and only if C' can be obtained from C by means of a flip. For example, Fig. 2 shows the transition graph $\mathbb{G}_5$. We note that $\mathbb{G}_n$ is acyclic and contains $(n - 1)!/2$ vertices. For more results related to plane geometric object transformation or graph transformation, the reader can refer to [1–9, 11, 13].

2. Preliminaries

We start with introducing some terminologies for convex cycles. From now on, unless otherwise mentioned, all arithmetics are taken modulo n . Recall that S is a set of n points arranged in convex position. We denote $ch(S)$ as the boundary of the convex hull of S . Let $C \in \mathcal{C}_n$ be a convex cycle. An edge $(u, v) \in C$ is said to be an *arc* if it lies on $ch(S)$, and a *chord* otherwise. For each edge $e = (u, v) \in C$, the minimum difference of the labels of u and v , denoted $\pi(e) = \min\{v - u, u - v\}$, is called the *depth* of e . The *barrier* of C , de-


 Figure 2: The transition graph $\mathbb{G}_5$.

noted as $\beta(C)$, is the number of chords that exist in C . Define $\mathcal{C}_n^k = \{C \in \mathcal{C}_n : \beta(C) = k\}$ for $k = 0, 1, \dots, n$. Clearly, $\mathcal{C}_n^0 = \{I_n\}$ and $\{\mathcal{C}_n^0, \mathcal{C}_n^1, \dots, \mathcal{C}_n^n\}$ forms a partition of $\mathcal{C}_n$. In addition, a convex cycle $C \in \mathcal{C}_n$ is said to be *eccentric* provided $\delta(C) = \delta(n)$.

Proposition 1 *For every convex cycle $C \in \mathcal{C}_n \setminus \{I_n\}$, there exists a convex cycle $C' \in \mathcal{C}_n$ with $\beta(C') < \beta(C)$ such that C can be transformed into C' by means of a flip.*

Proof. For convenience, we regard C as a directed cycle with a fixed orientation. Let $e = (u, v)$ be a chord with the minimum depth $\pi(e) = m \geq 2$ in C . By symmetry, we may suppose that u is the tail of e and

$v = u + m$ is the head of e . Let $w = u + 1$ and e' be the edge with tail at w . From the minimality of the depth of e , both e and e' cross each other. Thus, if we apply a flip on the pair $\{e, e'\}$, the arc (u, w) is produced in the newly created cycle C' . This shows that $\beta(C') < \beta(C)$. $\square$

Proposition 2 $\mathcal{C}_n^1 = \emptyset$.

Proof. Suppose on the contrary that there exists a convex cycle $C \in \mathcal{C}_n^1$ and let $e = (u, v)$ be the unique chord in C . Clearly, the set of points $S \setminus \{u, v\}$ is separated into different sides of e in the plane. If we traverse C along a fixed orientation, there exists at least one

edge, say e' , that connects two points in the different sides of e . This shows that e' is also a chord, a contradiction. $\square$

Proposition 3 *For every convex cycle $C \in \mathcal{C}_n \setminus \{I_n\}$, $\delta(C) \leq \beta(C) - 1$.*

Proof. From Proposition 2, we suppose that $C \in \mathcal{C}_n^k$ for $2 \leq k \leq n$. Proposition 1 shows that C can be transformed into I_n within k flips, i.e., $\delta(C) \leq \beta(C) = k$. Moreover, Proposition 2 shows that the last flip in the transformation certainly replaces two chords with two arcs. Thus, the result follows. $\square$

Proposition 4 *For $n \geq 7$, if $C \in \mathcal{C}_n$ is an eccentric convex cycle, then $C \in \mathcal{C}_n^n \cup \mathcal{C}_n^{n-1}$.*

Proof. Let C be an eccentric convex cycle in $\mathcal{C}_n$. By Theorem 2, $\delta(C) = \delta(n) \geq n - 2$. Suppose on the contrary that $C \in \mathcal{C}_n^k$ for $k \leq n - 2$, i.e., C contains at most $n - 2$ chords. Then, it follows from Proposition 3 that $\delta(C) \leq \beta(C) - 1 \leq n - 3$, a contradiction. $\square$

Next, we define some particular convex cycles and operations which are useful to achieve our proof. For each integer k relatively prime to n , we denote $C_{n,k}$ as the convex cycle that traverses every k steps in the cyclic order of S . Two convex cycles C and C' are said to be *rotationally isomorphic* (or simply say *isomorphic*), denoted by $C \cong C'$, if they look the same when one is turned a certain amount of rotations without concerning about the labels of points. For instance, all convex cycles in $\mathcal{C}_5^2$ (respectively, in $\mathcal{C}_5^3$) are rotationally isomorphic (see Fig. 2). Let $C \in \mathcal{C}_n$ be a convex cycle and (u, v) be an edge of C . By *subdividing* (u, v) we mean that (u, v) is removed from C and a new point is involved to join u and v . For integer $w = 1, 2, \dots, n$, we denote $C' = C \bullet (u, v)_w$ to mean that a convex cycle

$C' \in \mathcal{C}_{n+1}$ is obtained from C by subdividing (u, v) , and such that every point with label $\geq w$ augments its label by one and the new point of the subdivision receives w as its label to place in the cyclic order. In this way, we say that C' is obtained from C by *expanding* a point w to the edge (u, v) , or simply say that C' is an *expansion* of C . For example, Fig. 3 shows the convex cycle $C_{5,2}$ and all of its expansions obviating isomorphic ones. For convenience, we denote the three expansions by H_1, H_2 and H_3 , respectively.

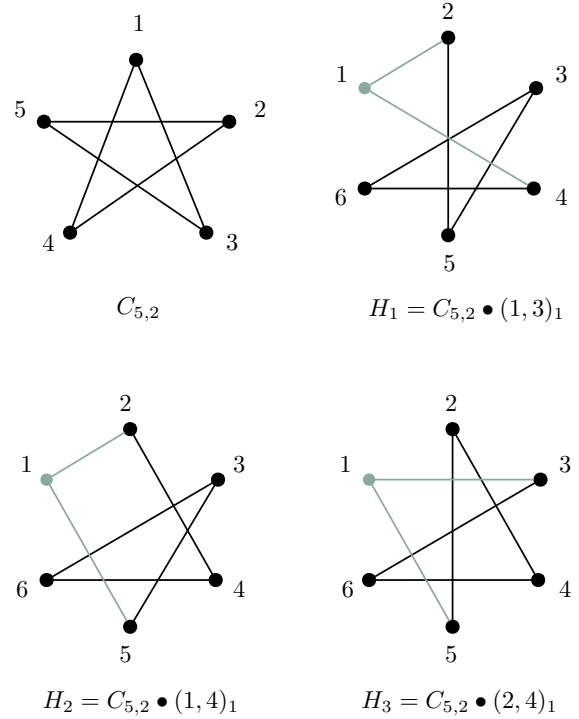
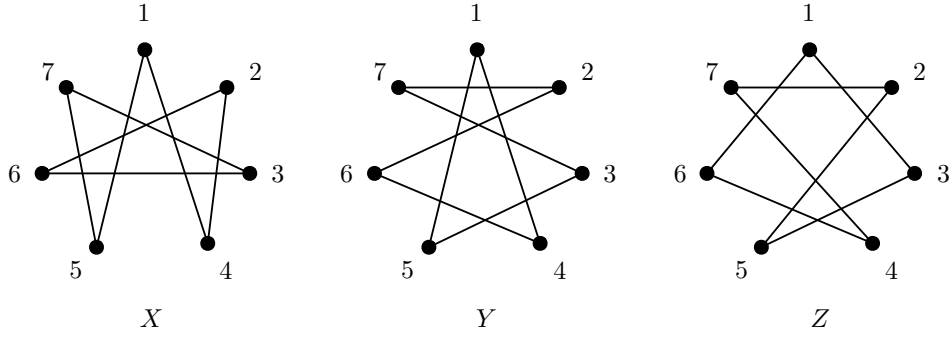
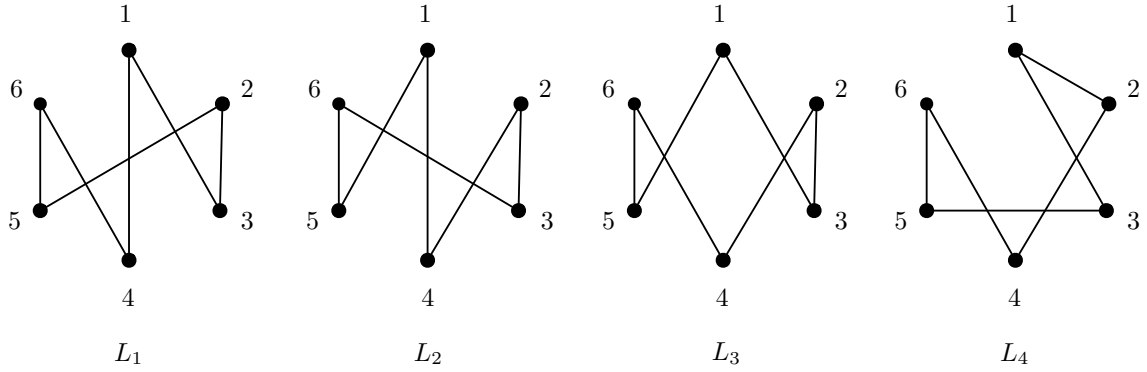


Figure 3: The convex cycle $C_{5,2}$ and three expansions of $C_{5,2}$.

The following two propositions directly follow from the definition of cycle expansion.

Proposition 5 *Let $C' = C \bullet (u, v)_w$. If $e = (u, v) \in C$ is an arc with $v = u + 1$, then*


 Figure 4: Three convex cycles in $\mathcal{C}_7^7$.

 Figure 5: four convex cycles in $\mathcal{C}_6^4$.

$$\beta(C') = \begin{cases} \beta(C), & \text{if } w = v; \\ \beta(C) + 1, & \text{if } w \in \{u, v + 1\} \text{ and } (w - 1, w) \notin C; \\ \beta(C) + 3, & \text{if } w \notin \{u, v, v + 1\} \text{ and } (w - 1, w) \in C; \\ \beta(C) + 2, & \text{otherwise.} \end{cases}$$

Proposition 6 Let $C' = C \bullet (u, v)_w$. If $e = (u, v) \in C$ is a chord with $v = u + \pi(e)$ where $\pi(e) \geq 2$, then

$$\beta(C') = \begin{cases} \beta(C), & \text{if } w \in \{u, u + 1, v, v + 1\} \\ & \text{and } (w - 1, w) \notin C; \\ \beta(C) + 2, & \text{if } w \notin \{u, u + 1, v, v + 1\} \\ & \text{and } (w - 1, w) \in C; \\ \beta(C) + 1, & \text{otherwise.} \end{cases}$$

We close this section with the following drawings of convex cycles in $\mathcal{C}_7^7$ (see Fig. 4) and $\mathcal{C}_6^4$ (see Fig. 5).

3. Proof of the main result

Since every convex cycle of order n can be obtained from a convex cycle of order $n - 1$ by a cycle expansion, we can gather up convex cycles of $\mathcal{C}_n^k$ by the rules of Propositions 5 and 6. In the following three lemmas, we assume that a convex cycle C is acquired from a convex cycle C' by an expansion.

Lemma 1 If $C \in \mathcal{C}_6^6$, then $C \cong H_3$.

Proof. By Propositions 5 and 6, either $C' \in \mathcal{C}_5^5$ or $C' \in \mathcal{C}_5^3$. Note that $\mathcal{C}_5^4 = \emptyset$ (see Fig. 2). Recall that $C_{5,2} \equiv C_{5,3}$ is the only member of $\mathcal{C}_5^5$ and we have shown in Fig. 3 that only convex cycles isomorphic to H_3 are expansions of $C_{5,2}$ with 6 chords. Since all convex cycles in $\mathcal{C}_5^3$ are isomorphic, we may consider the particular one $C' = (1, 3, 2, 5, 4)$ for the cycle

expansion. Clearly, $C' \bullet (2, 3)_5 \cong C' \bullet (4, 5)_3 \cong H_3$ and this is the only case of convex cycles with 6 chords for all possible expansions. $\square$

Lemma 2 *If $C \in \mathcal{C}_6^5$, then $C \cong H_1$ or $C \cong H_2$.*

Proof. By Propositions 5 and 6, C' must be contained in $\mathcal{C}_5^5$, $\mathcal{C}_5^3$ or $\mathcal{C}_5^2$. For $C' \in \mathcal{C}_5^5$, the expansion has been illustrated in Fig. 3. For $C' \in \mathcal{C}_5^3$, we may consider $C' = (1, 3, 2, 5, 4)$. It is easy to verify that $C' \bullet (2, 3)_1 \cong C' \bullet (4, 5)_2 \cong H_1$ and $C' \bullet (1, 3)_5 \cong C' \bullet (1, 4)_3 \cong H_2$. Moreover, all other expansions in this case cannot contain 5 chords. For $C' \in \mathcal{C}_5^2$, we may consider $C' = (1, 2, 4, 3, 5)$. Then, $C' \bullet (1, 2)_4 \cong C' \bullet (1, 5)_4 \cong H_1$ and $C' \bullet (3, 4)_1 \cong C' \bullet (3, 4)_4 \cong H_2$. Again, all other expansions in this case cannot contain 5 chords. $\square$

Lemma 3 *If $C \in \mathcal{C}_6^4$, then $C \cong L_i$ where $i \in \{1, 2, 3, 4\}$.*

Proof. By Propositions 5 and 6, either $C' \in \mathcal{C}_5^3$ or $C' \in \mathcal{C}_5^2$. For $C' \in \mathcal{C}_5^3$, we consider $C' = (1, 3, 2, 5, 4)$ and check the following:

- (i) $C' \bullet (4, 5)_1 \cong C' \bullet (2, 5)_5 \cong L_1$;
- (ii) $C' \bullet (2, 3)_2 \cong C' \bullet (2, 5)_3 \cong L_2$;
- (iii) $C' \bullet (2, 5)_4 \cong L_3$; and
- (iv) $C' \bullet (4, 5)_4 \cong C' \bullet (2, 3)_4 \cong C' \bullet (1, 3)_3 \cong C' \bullet (1, 4)_5 \cong L_2$.

Moreover, all other expansions in this case cannot contain 4 chords.

For $C' \in \mathcal{C}_5^2$, we consider $C' = (1, 2, 4, 3, 5)$ and check the following:

- (i) $C' \bullet (1, 2)_4 \cong L_1$;
- (ii) $C' \bullet (1, 5)_3 \cong L_2$;
- (iii) $C' \bullet (1, 2)_5 \cong C' \bullet (1, 5)_2 \cong L_3$; and
- (iv) $C' \bullet (2, 4)_5 \cong C' \bullet (3, 5)_2 \cong L_4$.

Again, all other expansions in this case cannot contain 4 chords. $\square$

Lemma 4 *Let C be a convex cycle in $\mathcal{C}_7^7$. Then, one of the following isomorphism holds:*

$$C \cong C_{7,2}, C \cong C_{7,3}, C \cong X, C \cong Y, \text{ or } C \cong Z.$$

where X, Y, Z are shown in Fig. 4, respectively.

Proof. Suppose that C is an expansion of a convex cycle $C' \in \mathcal{C}_6$. By Propositions 5 and 6, $C' \in \mathcal{C}_6^6 \cup \mathcal{C}_6^5 \cup \mathcal{C}_6^4$. Also, from Lemmas 1, 2 and 3, we only need to consider $C' = H_i$ for $i = 1, 2, 3$ or $C' = L_j$ for $j = 1, 2, 3, 4$, and check all expansions of C' that contain 7 chords. We list all possible cases as follows.

Case 1: $C' = H_1$. In this case, we have

$$\begin{aligned} H_1 \bullet (1, 2)_5 &\cong C_{7,3}, \\ H_1 \bullet (1, 2)_3 &\cong H_1 \bullet (1, 2)_6 \cong X, \\ H_1 \bullet (3, 5)_2 &\cong H_1 \bullet (4, 6)_2 \cong Y, \text{ and} \\ H_1 \bullet (3, 6)_2 &\cong Z. \end{aligned}$$

Case 2: $C' = H_2$. In this case, we have

$$\begin{aligned} H_2 \bullet (1, 2)_5 &\cong X, \\ H_2 \bullet (1, 2)_4 &\cong H_2 \bullet (1, 2)_6 \cong Y, \\ H_2 \bullet (3, 5)_2 &\cong H_2 \bullet (4, 6)_2 \cong Z, \text{ and} \\ H_2 \bullet (3, 6)_2 &\cong C_{7,2}. \end{aligned}$$

Case 3: $C' = H_3$. In this case, we have

$$\begin{aligned} H_3 \bullet (1, 3)_5 &\cong H_3 \bullet (1, 5)_3 \cong H_3 \bullet (2, 4)_6 \cong H_3 \bullet (4, 6)_3 \cong X, \\ H_3 \bullet (1, 3)_4 &\cong H_3 \bullet (1, 5)_4 \cong H_3 \bullet (2, 4)_1 \cong H_3 \bullet (4, 6)_2 \cong Y, \text{ and} \\ H_3 \bullet (2, 5)_1 &\cong H_3 \bullet (2, 5)_4 \cong H_3 \bullet (3, 6)_2 \cong H_3 \bullet (3, 6)_5 \cong Z. \end{aligned}$$

Case 4: $C' = L_1$. In this case, we have

$$L_1 \bullet (2, 3)_6 \cong L_1 \bullet (5, 6)_3 \cong X.$$

Case 5: $C' = L_2$. In this case, we have

$$L_2 \bullet (2, 3)_6 \cong L_2 \bullet (5, 6)_3 \cong X.$$

Case 6: $C' = L_3$. In this case, we have

$$L_3 \bullet (2, 3)_6 \cong L_3 \bullet (5, 6)_3 \cong Y.$$

Case 7: $C' = L_4$. In this case, we have

$$L_4 \bullet (1, 2)_6 \cong L_4 \bullet (5, 6)_2 \cong Z. \quad \square$$

Lemma 5 $\delta(7) \leq 5$.

Proof. We examine eccentric convex cycles in $\mathcal{C}_7$ to show the bound. By Proposition 4, we only need to verify convex cycles in $\mathcal{C}_7^6$ and $\mathcal{C}_7^7$. Since Proposition 3 has shown that every convex cycle $C \in \mathcal{C}_7^6$ has switching number $\delta(C) \leq 5$, we proceed with the verification of eccentric convex cycles in $\mathcal{C}_7^7$ according to the cases of Lemma 4 as follows.

For $C_{7,2}$, we obtain $\delta(C_{7,2}) \leq 5$ by the following 5 flips:

$$\begin{aligned} C_{7,2} &= (1, 3, 5, 7, 2, 4, 6) \xrightarrow{(1,3),(2,7)} \\ &\quad (1, 7, 5, 3, 2, 4, 6) \xrightarrow{(2,4),(3,5)} \\ &\quad (1, 7, 5, 2, 3, 4, 6) \xrightarrow{(2,5),(4,6)} \\ &\quad (1, 7, 5, 4, 3, 2, 6) \xrightarrow{(2,6),(5,7)} \\ &\quad (1, 7, 2, 3, 4, 5, 6) \xrightarrow{(1,6),(2,7)} \\ &\quad (1, 2, 3, 4, 5, 6, 7) = I_7 \end{aligned}$$

For $C_{7,3}$, we obtain $\delta(C_{7,3}) \leq 5$ by the following 5 flips:

$$\begin{aligned} C_{7,3} &= (1, 4, 7, 3, 6, 2, 5) \xrightarrow{(1,4),(3,7)} \\ &\quad (1, 7, 4, 3, 6, 2, 5) \xrightarrow{(2,5),(3,6)} \\ &\quad (1, 7, 4, 3, 2, 6, 5) \xrightarrow{(1,5),(2,6)} \\ &\quad (1, 7, 4, 3, 2, 5, 6) \xrightarrow{(2,5),(4,7)} \\ &\quad (1, 7, 2, 3, 4, 5, 6) \xrightarrow{(1,6),(2,7)} \\ &\quad (1, 2, 3, 4, 5, 6, 7) = I_7 \end{aligned}$$

For X , we obtain $\delta(X) \leq 5$ by the following 5 flips:

$$\begin{aligned} X &= (1, 4, 2, 6, 3, 7, 5) \xrightarrow{(2,6),(3,7)} \\ &\quad (1, 4, 2, 3, 6, 7, 5) \xrightarrow{(2,4),(3,6)} \\ &\quad (1, 4, 3, 2, 6, 7, 5) \xrightarrow{(1,4),(2,6)} \\ &\quad (1, 2, 3, 4, 6, 7, 5) \xrightarrow{(1,5),(4,6)} \\ &\quad (1, 2, 3, 4, 5, 7, 6) \xrightarrow{(1,6),(5,7)} \\ &\quad (1, 2, 3, 4, 5, 6, 7) = I_7 \end{aligned}$$

For Y , we obtain $\delta(Y) \leq 4$ by the following 4 flips:

$$\begin{aligned} Y &= (1, 4, 6, 2, 7, 3, 5) \xrightarrow{(2,6),(3,7)} \\ &\quad (1, 4, 6, 7, 2, 3, 5) \xrightarrow{(3,5),(4,6)} \\ &\quad (1, 4, 3, 2, 7, 6, 5) \xrightarrow{(1,4),(2,7)} \\ &\quad (1, 2, 3, 4, 7, 6, 5) \xrightarrow{(1,5),(4,7)} \\ &\quad (1, 2, 3, 4, 5, 6, 7) = I_7 \end{aligned}$$

For Z , we obtain $\delta(Z) \leq 4$ by the following 4 flips:

$$\begin{aligned} Z &= (1, 3, 5, 2, 7, 4, 6) \xrightarrow{(3,5),(4,6)} \\ &\quad (1, 3, 4, 7, 2, 5, 6) \xrightarrow{(1,3),(2,7)} \\ &\quad (1, 7, 4, 3, 2, 5, 6) \xrightarrow{(1,6),(4,7)} \\ &\quad (1, 7, 6, 5, 2, 3, 4) \xrightarrow{(1,4),(2,5)} \\ &\quad (1, 2, 3, 4, 5, 6, 7) = I_7 \end{aligned} \quad \square$$

Theorem 3 For any integer $n \geq 7$, $\delta(n) \leq n - 2$.

Proof. We prove the theorem by induction on n . Lemma 5 shows the base case $\delta(7) \leq 5$. Suppose that $n \geq 8$ and the theorem holds for integer no more than $n - 1$. Let $C \in \mathcal{C}_n$ be a convex cycle with $\delta(C) = \delta(n)$. By Proposition 1, at most one flip is required to produce

a new convex cycle C' containing an arc e absent from C . Since $\beta(C') < \beta(C)$, the convex cycle C can be transformed into I_n in at most $\delta(C') + 1$ flips. We then contract the edge e in C' and denote the resulting cycle with $n - 1$ points by $\hat{C}$. Clearly, $\delta(\hat{C}) = \delta(C')$. From the induction hypothesis, it holds that $\delta(\hat{C}) \leq n - 3$. Therefore, we have

$$\begin{aligned}\delta(n) = \delta(C) &\leq \delta(C') + 1 = \delta(\hat{C}) + 1 \\ &\leq (n - 3) + 1 = n - 2.\end{aligned}$$

□

As a consequence, the correctness of the following corollary directly follows from Theorems 2 and 3.

Corollary 4 *For any integer $n \geq 7$, $\delta(n) = n - 2$.*

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